Torus Solutions Beyond GRW: verification notebook

Generated by Claude (Anthropic) to accompany the supplement "Torus Solutions Beyond the GRW Assumption". Bud Simrin's part was the series of questions that led to it. He did not check the derivations himself, but he ran this notebook, and its outputs match the supplement. This notebook checks the two main claims of the supplement for two curled circles (the 6D_1 setting, with k = 0):

1. The spatial metric below is Ricci-flat exactly when $e4^2 + e4\,e5 + e5^2 = 1$, which is the same as $s^2 + e4^2 + e5^2 = 2$ with $s = e4 + e5$.

2. Its curvature (the Kretschmann invariant) stays finite at r = 2m only at (e4, e5) = (1, 0) and (0, 1), the GRW solutions of the 6D_1 notebook.

Evaluate the cells from top to bottom. The Riemann tensor step takes up to a minute or so.

Metric

Spatial metric h, with f = 1 – 2m/r (so $\chi = -2m$) and s = e4 + e5:

In[3882]:=

```
ClearAll["Global`*"];
coords = {r, θ, ϕ, ψ4, ψ5};
f = 1 - 2 m / r; s = e4 + e5;
g = DiagonalMatrix[
    {f^-s, f^(1 - s) r^2, f^(1 - s) r^2 Sin[θ]^2, f^e4 p4^2, f^e5 p5^2}];
gi = Inverse[g]; n = Length[coords];
g // MatrixForm
```

Out[3887]//MatrixForm=

$$\begin{pmatrix} \left(1-\frac{2m}{r}\right)^{-e4-e5} & 0 & 0 & 0 & 0 \\ 0 & \left(1-\frac{2m}{r}\right)^{1-e4-e5} r^2 & 0 & 0 & 0 \\ 0 & 0 & \left(1-\frac{2m}{r}\right)^{1-e4-e5} r^2 \operatorname{Sin}[\theta]^2 & 0 & 0 \\ 0 & 0 & 0 & p4^2 \left(1-\frac{2m}{r}\right)^{e4} & 0 \\ 0 & 0 & 0 & 0 & p5^2 \left(1-\frac{2m}{r}\right)^{e5} \end{pmatrix}$$

Christoffel symbols and Ricci tensor

In[3888]:=
```
Γ = Table[
    Sum[gi[[a, d]] (D[g[[d, b]], coords[[c]]] + D[g[[d, c]], coords[[b]]] - D[g[[b, c]], coords[[d]]]),
      {d, n}] / 2, {a, n}, {b, n}, {c, n}];
ric = Table[Sum[D[Γ[[a, b, c]], coords[[a]]] - D[Γ[[a, b, a]], coords[[c]]] + Sum[
       Γ[[a, a, d]] × Γ[[d, b, c]] - Γ[[a, c, d]] × Γ[[d, b, a]], {d, n}], {a, n}], {b, n}, {c, n}];
```

Off-diagonal components (all zero) and the diagonal components:

In[3890]:=
```
{Union[Flatten[Simplify[ric - DiagonalMatrix[Diagonal[ric]]]]],
 Factor /@ FullSimplify[Diagonal[ric], m > 0 && r > 2 m]}
```

Out[3890]=
$$\left\{\{0\}, \left\{-\frac{2\left(-1+e4^2+e4\,e5+e5^2\right)m^2}{(2m-r)^2 r^2}, 0, 0, 0, 0\right\}\right\}$$

Only R_11 (the radial component; index 1 is r) is nonzero, and it vanishes exactly when $e4^2 + e4\,e5 + e5^2 = 1$.

The GRW (s = 1) solutions are special cases

With e5 = 1 – e4 (so s = 1), the condition becomes e4(e4 – 1) = 0:

In[3891]:=
```
Factor[(e4^2 + e4 e5 + e5^2 - 1) /. e5 → 1 - e4]
```

Out[3891]=
$$(-1+e4)\,e4$$

Kretschmann invariant

In[3892]:=
```
riem = Table[D[Γ[[a, b, d]], coords[[c]]] - D[Γ[[a, b, c]], coords[[d]]] +
     Sum[Γ[[a, c, k]] × Γ[[k, b, d]] - Γ[[a, d, k]] × Γ[[k, b, c]], {k, n}],
    {a, n}, {b, n}, {c, n}, {d, n}];
low = Table[Sum[g[[a, k]] × riem[[k, b, c, d]], {k, n}], {a, n}, {b, n}, {c, n}, {d, n}];
up =
  Table[gi[[b, b]] × gi[[c, c]] × gi[[d, d]] × riem[[a, b, c, d]], {a, n}, {b, n}, {c, n}, {d, n}];
K = FullSimplify[Total[Flatten[low up]], m > 0 && r > 2 m]
```

Out[3895]=

$$\frac{1}{r^4\,(-2\,m+r)^4}\,4\,m^2\left(1-\frac{2\,m}{r}\right)^{2\,(e4+e5)}\Big(\big(7\,e4^4+4\,e4^3\,(4+3\,e5)+4\,e4\,(1+e5)\,(2+e5\,(2+3\,e5))+(1+e5)^2\,(3+e5\,(2+7\,e5))+e4^2\,(14+e5\,(20+13\,e5))\big)\,m^2-4\,\big(4\,e4^3+2\,e5\,(1+e5)\,(1+2\,e5)+e4^2\,(6+5\,e5)+e4\,(2+e5\,(6+5\,e5))\big)\,m\,r+12\,\left(e4^2+e4\,e5+e5^2\right)\,r^2\Big)$$

For the 6D_1 GRW solution (e4, e5) = (1, 0) this is the Schwarzschild value $48m^2/r^6$, finite at r = 2m:

In[3896]:=
```
FullSimplify[K /. {e4 → 1, e5 → 0}, m > 0 && r > 2 m]
```

Out[3896]=

$$\frac{48\,m^2}{r^6}$$

In general K is (1 – 2m/r)^(2s) / (r – 2m)^4 times a polynomial. Points on the solution curve where that polynomial vanishes at r = 2m (the only places where K can stay finite):

In[3897]:=
```
poly = Numerator[Together[K r^4 (r - 2 m)^4 / (4 m^2 (1 - 2 m / r)^(2 (e4 + e5)))]];
Solve[{(poly /. r → 2 m) == 0, e4^2 + e4 e5 + e5^2 == 1}, {e4, e5}, Reals]
```

Out[3898]=
```
{{e4 → 0, e5 → 1}, {e4 → 1, e5 → 0}}
```

Numerical values of K approaching r = 2m (with m = 1/2) at four points on the solution curve:

In[3899]:=

```
pt[s0_] := Module[{d = Sqrt[s0^2/4 - (s0^2 - 1)]}, {s0/2 + d, s0/2 - d}];
TableForm[Table[Module[{e = pt[s0]}, Join[{s0, N[e, 3]},
    Table[ScientificForm[N[K /. {e4 → e[[1]], e5 → e[[2]], m → 1/2, r → 1 + eps}], 2],
     {eps, {1/10, 1/100, 1/1000, 1/10000}}]]], {s0, {1, 11/10, 3/5, -1/2}}],
 TableHeadings → {None, {"s", "(e4, e5)", "r-2m = 0.1", "0.01", "0.001", "0.0001"}}]
```

Out[3900]//TableForm=

s	(e4, e5)	r−2m = 0.1	0.01	0.001	0.0001
1	1.00 0	6.8	1.1×10^{1}	1.2×10^{1}	1.2×10^{1}
$\frac{11}{10}$	0.854 0.246	5.8	1.1×10^{2}	6.8×10^{3}	4.3×10^{5}
$\frac{3}{5}$	1.15 −0.554	3.9×10^{2}	2.5×10^{5}	1.6×10^{8}	$1.\times10^{11}$
$-\frac{1}{2}$	0.651 −1.15	8.6×10^{5}	9.2×10^{10}	9.3×10^{15}	9.3×10^{20}

Only the s = 1 row stays finite; the others grow without bound. The point r = 2m lies at a finite proper distance (the integral of f^(–s/2) dr converges there), so these are genuine curvature singularities.